Comparing Specialized and General-Purpose LLMs for Mental Health Support: A Longitudinal Analysis of Conversational Behaviors During a Month-Long DBT Skills Training Study

Abstract

Large language models (LLMs) are increasingly used for mental health support, yet general-purpose systems often lack domain-specific knowledge and therapeutic structure. Retrieval-augmented generation (RAG) offers a promising approach for grounding conversational AI in evidence-based mental health resources, but little is known about how such systems differ from general-purpose LLMs in supporting users over time. This study compares an RAG-based mental health conversational AI (DBT-AI) with a *general-purpose chatbot* for Dialectical Behavior Therapy (DBT) skills training. Through a one-month field study with 30 participants, we collected longitudinal conversation data and analyzed conversational behaviors using a six-dimensional framework. Our findings suggest that positive mental health support may rely less on emotional validation alone and more on facilitating structured reflection and actionable coping strategies. The DBT-AI demonstrated greater consistency in delivering skills-focused and therapeutically grounded support. By providing an interpretable analysis of conversational behaviors, this work contributes insights into the design and evaluation of trustworthy AI-enabled mental health interventions and highlights opportunities for responsible deployment of LLM-based mental health technologies.

CCS Concepts

• **Applied computing** → **Consumer health**; **Psychology**; • **Human-centered computing** → **Empirical studies in HCI**; **Natural language interfaces**; • **Computing methodologies** → **Natural language processing**.

Keywords

mental health chatbot, large language models, conversational behavior, emotional improvement, personalized interventions, digital mental health, AI-supported mental health

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1 Introduction

Recent research has demonstrated the potential of conversational AI to support mental health screening, therapeutic skill practice, and ongoing self-management through mobile and everyday computing platforms [27, 28]. The emergence of large language models (LLMs) has further expanded these capabilities by enabling more personalized, context-aware, and scalable interactions [4, 13]. As a result, many individuals are increasingly turning to general-purpose LLM applications for emotional support, coping advice, and psychological skills training [10, 32]. In parallel, specialized mental health systems are being developed that augment LLMs with curated clinical knowledge and evidence-based therapeutic frameworks, often through techniques such as retrieval-augmented generation (RAG), with the goal of providing more reliable and therapeutically grounded support.

Prior research has demonstrated the potential of LLM-based mental health chatbots to improve user engagement and facilitate evidence-based therapeutic interventions [20, 35]. However, comparatively little is known about how specialized mental health systems differ from general-purpose LLMs in practice. Existing evaluations primarily focus on outcomes such as symptom reduction, user satisfaction, or emotional improvement [9, 35]. While these studies demonstrate effectiveness, they provide limited insight into the conversational strategies through which support is delivered. Consequently, our understanding of how different AI systems facilitate reflection, coping, and skills practice remains limited [16, 18, 23].

Understanding these conversational differences is particularly important as generative AI becomes increasingly integrated into mobile and ubiquitous mental health technologies. Unlike traditional digital interventions that deliver static content, LLM-powered systems can provide adaptive, personalized, and conversational support throughout users' daily lives [4, 13]. At the same time, the growing deployment of AI-powered mental health applications raises important questions regarding the consistency, quality, and therapeutic characteristics of AI-generated support in real-world settings [9, 12]. As conversational AI becomes embedded in mobile mental health applications, it is important to understand how different AI architectures shape user experiences and therapeutic interactions during everyday use. Examining conversational behaviors in deployed systems can inform the design of future ubiquitous mental health technologies that are both scalable and therapeutically meaningful.

Therefore, we compared a general-purpose LLM with a specialized RAG-based LLM embedded in a mobile chatbot for Dialectical Behavior Therapy (DBT) skills training. Through a month-long field deployment with 30 participants, we collected longitudinal conversational data and analyzed the interactions using a six-dimensional

framework adapted from prior research on conversational behaviors in AI-supported mental health interventions. Our findings suggest that effective mental health conversations rely less on emotional validation alone and more on facilitating structured reflection and actionable coping strategies [24, 33]. Compared with the *general-purpose chatbot*, the RAG-based system consistently delivered more skills-focused, actionable, and therapeutically grounded support, highlighting the potential of domain-specific LLMs for mental health interventions.

This paper makes three contributions. First, we provide empirical evidence of how conversational behaviors differ between a specialized RAG-based mental health system and a *general-purpose chatbot* during a month-long, real-world deployment. Second, we propose a six-dimensional framework for characterizing conversational behaviors in AI-supported mental health interventions. Third, we derive design implications for developing mobile and ubiquitous AI-supported mental health interventions.

2 Study Design

The data analyzed in this study were collected from a 30-day deployment of two web-based mental health intervention applications with university students. The baseline application integrated a *general-purpose chatbot* that provided emotional support through open-ended conversations. In contrast, the experimental application employed a specialized *DBT-AI chatbot* that combined Dialectical Behavior Therapy (DBT) principles, retrieval-augmented generation (RAG), and structured skills training to deliver evidence-based therapeutic guidance [19, 22].

All conversations were anonymized prior to analysis. The study was conducted with institutional ethics approval; participants provided informed consent. Conversational data were stored on access-controlled servers. The final dataset consisted of 159 conversation sessions from the *general-purpose chatbot* and 208 conversation sessions from the *DBT-AI chatbot*. Each session contained user messages, chatbot responses, timestamps, and corresponding sentiment measurements. By analyzing data collected from a month-long deployment, this study captures conversational behaviors emerging during everyday interactions with AI-supported mental health technologies.

2.1 Comparison Systems

Both systems shared the same LLM-based conversational backbone, enabling free-form natural-language dialogue in a supportive and non-judgmental tone. The ***general-purpose chatbot*** relied solely on this backbone and provided open-ended emotional support through a generic supportive prompt, without following any structured therapeutic framework.

The ***DBT-AI chatbot*** extended the same backbone with three therapeutic components: (1) DBT-specific prompting to guide responses around evidence-based DBT skills (e.g., mindfulness, distress tolerance), (2) retrieval-augmented generation (RAG) grounded in a curated DBT knowledge base to improve therapeutic consistency and reduce hallucinations [19], and (3) a structured interaction workflow that guided users through skill identification, contextual reflection, personalized practice, and feedback [22]. Figure 1 illustrates this workflow in practice. Before initiating mindfulness

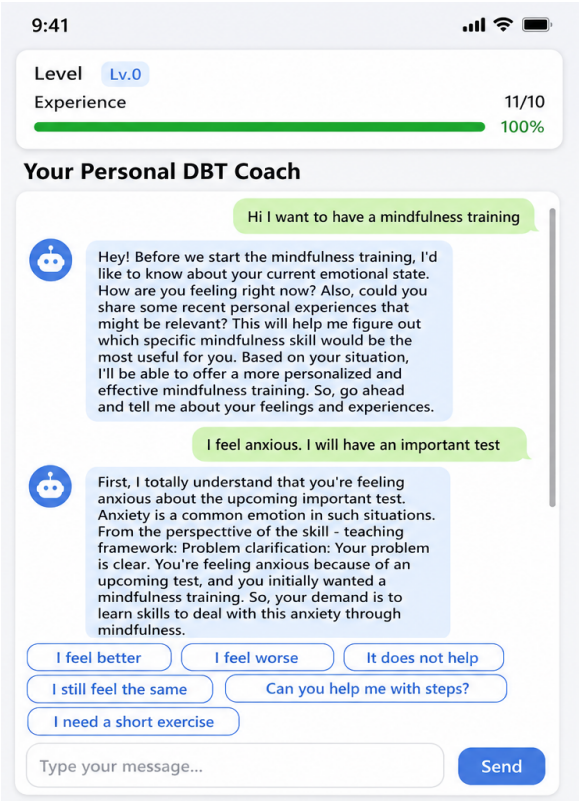


Figure 1: Conversational interface of the *DBT-AI chatbot*.

training, the system first elicits contextual information about the user's emotional state and situation, then structures subsequent responses around a skill-teaching framework rather than providing generic supportive replies. This design allows us to examine how therapeutic grounding and structured guidance influence conversational support in real-world use.

2.2 Conversational Behavior Framework

To characterize conversational behaviors relevant to DBT-based digital mental health support, we developed a six-dimensional conversational behavior framework based on literature from counseling psychology and motivational interviewing. Because our goal was to characterize differences between a *general-purpose chatbot* and a DBT-based conversational system, the framework focuses on behaviors related to emotional support, reflection, and therapeutic skill delivery rather than on providing a comprehensive evaluation framework for mental health chatbots. The framework includes the following six dimensions:

- **Emotional Attunement**: the extent to which the chatbot recognizes, validates, and responds appropriately to the user's emotional state [29, 31].
- **Contextual Fit**: the degree to which responses are relevant to the user's situation, concerns, and conversational context [31, 32].

- **Guidance Clarity**: the clarity and comprehensibility of explanations, coping strategies, and therapeutic guidance provided by the chatbot [7, 8].
- **Actionability**: the extent to which the chatbot offers concrete and feasible actions that users can apply in their daily lives [3, 22].
- **Reflective Facilitation**: the degree to which the chatbot encourages self-reflection, self-awareness, and deeper exploration of thoughts and emotions [24].
- **Safety & Ethics**: the appropriateness, safety, and ethical quality of the chatbot's responses [30].

These dimensions capture both supportive and instructional aspects of conversational mental health support while remaining interpretable for behavioral analysis.

2.3 Conversational Behavior Evaluation

Each conversation session was evaluated using an LLM-as-a-Judge approach, which provides a scalable method for assessing large volumes of open-ended conversational interactions [14, 34]. For every session, the evaluator assigned scores ranging from 1 (very low) to 5 (very high) on each of the six dimensions. To assess the reliability of automated ratings, we randomly sampled 10% of the conversation sessions for manual validation. One researcher independently re-scored each sampled session using the proposed framework. Agreement between human and automated ratings was assessed using quadratic weighted kappa (QWK) [5]. All substantive dimensions demonstrated substantial agreement, with QWK values ranging from 0.65 to 0.84 (excluding Safety & Ethics, where automated scores were near-ceiling and QWK was not informative), indicating that the framework could be reliably applied to large-scale conversational analysis.

2.4 Emotional Improvement Measure

We first examined whether grounding conversational support in DBT knowledge and structured guidance translated into greater emotional improvement. Analyses were conducted at the session level ($N_{\text{GP}} = 159$, $N_{\text{DBT}} = 208$). To quantify emotional change, we measured both model-derived sentiment trajectories and self-reported mood changes for each conversation session. Session-level sentiment was estimated using a multilingual transformer-based sentiment analysis model [2], which assigned each user message a sentiment score ranging from 0 to 10, with higher scores indicating more positive emotional states. Emotional improvement ($\Delta Sentiment$) was calculated as the difference between the sentiment scores of the final and initial user messages within a session. Positive values indicate emotional improvement, whereas negative values indicate emotional deterioration.

To complement this objective measure, we also collected self-reported mood ratings through an emoji-based mood tracker, which were mapped to the same 0–10 scale. Self-reported mood change ($\Delta Mood$) was computed as the difference between participants' post-session and pre-session ratings, providing a lightweight indicator of perceived emotional change during everyday interactions with the conversational mental health systems.

Table 1: Session-level emotional outcomes (*general-purpose chatbot* vs. *DBT-AI chatbot*). Bold marks a statistically significant advantage for the DBT-informed system.

Metric	General-purpose	DBT-informed
$\Delta Mood$ (M)	0.63	0.73
$\Delta Sentiment$ (M)	0.30	**1.28**
Sentiment improve rate	32.7%	**51.5%**

2.5 Analysis

We conducted two sets of analyses. First, we compared conversational behavior scores between the *general-purpose chatbot* and the *DBT-AI chatbot* to identify behavioral differences. Second, we examined the relationship between conversational behaviors and emotional improvement. Correlation and regression analyses were performed to identify which conversational behaviors were most strongly associated with positive emotional outcomes. This analysis enables us to connect conversational processes with user emotional experiences and better understand how different AI architectures shape mental health support in real-world settings.

3 Results

3.1 Reliability of Automated Behavior Scoring

Before examining conversational behaviors across the month-long deployment, we evaluated the reliability of automated scoring. On the 10% of sessions independently re-scored by a human rater, automated and human ratings showed substantial agreement across all substantive dimensions, with quadratic weighted kappa (QWK) values ranging from 0.65 to 0.84 [17]. Safety & Ethics was excluded from this comparison because both raters assigned near-ceiling scores to nearly every session, leaving insufficient variance for QWK to be informative [6]. These results suggest that the proposed framework can reliably support large-scale behavioral analysis of deployed conversational mental health systems.

3.2 Emotional Improvement

As shown in Table 1, the *DBT-AI chatbot* achieved substantially greater text-based emotional improvement than the *general-purpose chatbot* ($M = 1.28$ vs. 0.30; $p < .001$, Cohen's $d = 0.38$), and a larger proportion of sessions showed positive sentiment change (51.5% vs. 32.7%; $p < .001$). Self-reported mood improved in both conditions, but the gap was smaller and less pronounced ($M = 0.73$ vs. 0.63), indicating that the DBT-informed system's advantage was expressed most clearly in users' spontaneous language rather than in coarse self-report [15]. These findings suggest that grounding conversational support in therapeutic knowledge can improve short-term emotional outcomes during real-world interactions.

3.3 Differences in Conversational Behavior

Having established differences in emotional outcomes, we next examined how the two AI architectures differed in their conversational behaviors. As Table 2 shows, rather than producing uniformly higher-quality conversations, the two systems exhibited distinct

Table 2: Mean conversational behavior scores (1–5) by system (*general-purpose chatbot* vs. *DBT-AI chatbot*). Bold marks a statistically significant between-condition difference.

Dimension	General-purpose	DBT-informed
Emotional Attunement	**4.24**	3.71
Contextual Fit	4.04	3.85
Guidance Clarity	4.05	**4.51**
Actionability	4.17	**4.53**
Reflective Facilitation	3.40	**3.74**
Safety & Ethics	5.00	4.98
Average score	4.15	4.22

conversational profiles.: the systems did not differ in mean composite score ($M = 4.22$ vs. 4.15, $p = .57$), yet they diverged sharply in the *type* of support they provided.

The *DBT-AI chatbot* scored significantly higher on Guidance Clarity ($M = 4.51$ vs. 4.05, $p < .001$, $d = 0.46$), Actionability ($M = 4.53$ vs. 4.17, $p = .001$, $d = 0.38$), and Reflective Facilitation ($M = 3.74$ vs. 3.40, $p < .001$, $d = 0.36$). The *general-purpose chatbot*, in contrast, scored markedly higher on Emotional Attunement ($M = 4.24$ vs. 3.71, $p < .001$, $|d| = 0.73$)—the largest single between-condition difference in the study. Contextual Fit and Safety & Ethics did not differ significantly. These results indicate that the *DBT-AI chatbot* emphasized structured guidance, concrete coping strategies, and self-reflection, whereas the *general-purpose chatbot* primarily emphasized emotional validation.

3.4 Conversational Behaviors Associated with Emotional Improvement

We then tested which behaviors were linked to emotional improvement by relating session-level behavior scores to session-level sentiment change.

Across the pooled dataset, the composite behavior score was positively associated with emotional improvement ($r = 0.20$–0.23 across improvement measures, all $p < .001$). At the dimension level, Guidance Clarity and Reflective Facilitation showed the strongest associations with $\Delta Sentiment$ ($r = 0.20$, $p < .001$), whereas Emotional Attunement was comparatively weak. Safety & Ethics again showed negligible variance and was uninformative.

Comparing improved and non-improved sessions yielded a consistent picture. Sessions with positive sentiment change received higher conversational behavior scores than those without ($M = 4.37$ vs. 4.09, $p < .001$, $d = 0.53$); the same held for mood-improved sessions ($M = 4.38$ vs. 4.18, $p = .015$, $d = 0.40$) and for overall-improved sessions ($M = 4.34$ vs. 4.09, $p < .001$, $d = 0.48$).

Crucially, this association was concentrated in the DBT-informed system. Conversation quality was significantly correlated with sentiment improvement for the *DBT-AI chatbot* ($r = 0.33$, $p < .001$, $n = 208$), but not for the *general-purpose chatbot* ($r = 0.04$, $p = .61$, $n = 159$; Figure 2). Conversational quality therefore became consequential only when interactions were organized around therapeutic skill delivery rather than open-ended supportive dialog. Taken together, the behaviors most strongly tied to emotional improvement

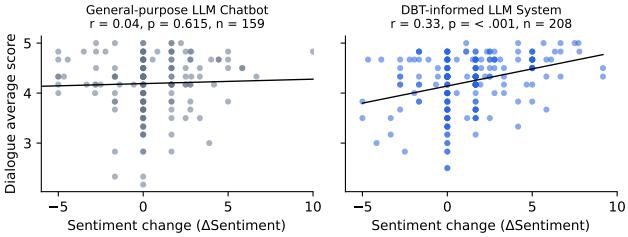


Figure 2: $\Delta Sentiment$ vs. dialogue average score, stratified by chatbot condition.

were not those reflecting emotional responsiveness, but those that helped users understand, reflect on, and act on their situations.

3.5 What Explains the *DBT-AI chatbot*'s Advantage?

To isolate the behaviors driving emotional improvement within each system, we estimated, separately for each chatbot condition, a linear regression of session-level sentiment change ($\Delta Sentiment$) on the six conversational behavior dimensions:

$$\begin{aligned}\Delta Sentiment_i = &\beta_0 + \beta_1 EmotionalAttunement_i + \beta_2 ContextualFit_i \\ &+ \beta_3 GuidanceClarity_i + \beta_4 Actionability_i \\ &+ \beta_5 ReflectiveFacilitation_i + \beta_6 SafetyEthics_i \\ &+ \epsilon_i\end{aligned} \tag{1}$$

where $\Delta Sentiment_i$ denotes the emotional improvement observed during session i, and the six predictors correspond to the conversational behavior dimensions.

Table 3: System-specific regression models predicting session-level emotional improvement ($\Delta Sentiment$); *general-purpose chatbot* vs. *DBT-AI chatbot*.

Predictor	General-purpose	DBT-informed
Emotional Attunement	0.778*	0.443
	($p = .047$)	($p = .097$)
Contextual Fit	-0.207	0.058
	($p = .801$)	
Guidance Clarity	0.108	1.396*
	($p = .747$)	($p = .014$)
Actionability	-0.050	-0.576
	($p = .878$)	($p = .261$)
Reflective Facilitation	-0.199	0.328
	($p = .560$)	($p = .115$)
Safety & Ethics	—	-1.942
	(no variance)	($p = .104$)
n	159	208
R^2	0.027	0.147
Adjusted R^2	-0.005	0.122

The two models behaved very differently (Table 3). For the *general-purpose chatbot* ($n = 159$), the model explained little variance in emotional improvement ($R^2 = 0.027$), and Emotional Attunement was the only significant predictor ($b = 0.778$, $p = .047$). For the *DBT-AI chatbot* ($n = 208$), the model explained considerably more variance ($R^2 = 0.147$), and Guidance Clarity was the only significant predictor ($b = 1.396$, $p = .014$): sessions in which the chatbot provided clearer explanations, more understandable coping strategies, and more structured guidance showed larger positive sentiment change. No other dimension reached significance in either model.

These results indicate that conversational behaviors operate differently across architectures. In open-ended supportive dialog, emotional attunement was the only behavior that weakly tracked improvement, and behaviors collectively explained little variance ($R^2 = 0.027$). In the structured DBT-based system, the clarity of guidance became the dominant correlate of positive outcomes ($R^2 = 0.147$), which is consistent with the design goals of skills-based interventions, in which improvement arises from helping users understand, practice, and apply coping strategies rather than from emotional validation alone [21, 26].

4 Discussion

4.1 Beyond Emotional Support: What Makes the *DBT-AI chatbot* Effective?

Previous work has shown that LLM-based mental health systems can improve user outcomes, but less is known about how different conversational designs contribute to these improvements. Rather than finding a general advantage for higher conversational quality, our results suggest that the effectiveness of AI-supported mental health conversations depends on how therapeutic support is structured and delivered.

Although the *general-purpose chatbot* demonstrated stronger Emotional Attunement, the *DBT-AI chatbot* produced greater emotional improvement through clearer guidance, more actionable recommendations, and stronger support for self-reflection. This suggests that emotional validation alone may be insufficient for promoting meaningful change during everyday mental health interactions. Instead, conversations that help users organize their experiences, understand coping strategies, and identify concrete next steps may provide greater therapeutic value.

More broadly, Guidance Clarity represents a behavioral characteristic that can be observed, evaluated, and potentially optimized during deployment. As conversational AI becomes increasingly embedded in mental health technologies, interpretable behavioral indicators may provide practical targets for improving system reliability, transparency, and therapeutic consistency.

4.2 Guidance as a Mechanism for Robust and Responsible Mental Health AI

Among the six conversational dimensions, Guidance Clarity was the only significant predictor of emotional improvement within the *DBT-AI chatbot*, whereas Emotional Attunement was the only significant predictor in the *general-purpose chatbot*. This difference suggests that the contribution of conversational behaviors depends on the underlying system architecture rather than representing universal properties of supportive dialogue.

Within the *DBT-AI chatbot*, clear guidance may help users translate therapeutic concepts into concrete coping strategies that can be applied beyond the conversation itself. This interpretation is consistent with skill-based psychotherapy, where improvement is achieved through learning and practicing adaptive behaviors [11].

More broadly, Guidance Clarity represents a behavioral characteristic that can be observed, evaluated, and potentially optimized during deployment. As conversational AI becomes increasingly embedded in mental health technologies, interpretable behavioral indicators may provide practical targets for improving system reliability, transparency, and therapeutic consistency.

4.3 Real-World Conversational Behaviors Are Context-Dependent

Our findings also suggest that conversational behaviors should be interpreted within the broader interaction context rather than in isolation. Although Actionability and Reflective Facilitation were associated with emotional improvement in bivariate analyses, neither remained significant after controlling for other conversational dimensions.

This pattern indicates that effective mental health conversations emerge from combinations of complementary behaviors rather than individual conversational features. For example, reflective questions may be more effective when accompanied by clear explanations, while actionable recommendations may depend on users first understanding why a particular strategy is relevant to their situation. This observation aligns with the Behavioral Intervention Technology (BIT) model, which emphasizes that intervention components operate as coordinated systems rather than independent design elements. [25].

For designers of AI-supported mental health applications, future conversational systems may therefore benefit from considering how multiple behavioral components work together throughout an ongoing interaction rather than optimizing individual response characteristics independently.

4.4 Limitations and Future Work

Several limitations should be considered. First, the study was conducted with university students, which may limit generalizability to broader, clinically diverse populations [1]. Second, the analyses are correlational and cannot establish causal relationships between conversational behaviors and emotional outcomes. Third, although automated conversational scoring demonstrated substantial agreement with manual validation on a 10% researcher-checked sample, only a subset of conversations was manually re-rated.

Future work could extend this behavioral framework from offline evaluation to real-time monitoring of conversational AI systems. Rather than using behavioral dimensions solely for post hoc analysis, indicators such as Guidance Clarity, Actionability, and Safety & Ethics could serve as online feedback signals for adaptive intervention strategies. Such behavioral monitoring may enable conversational mental health systems to become more transparent, accountable, and responsive during everyday use, supporting the responsible deployment of AI-enabled mental health technologies.

5 Conclusion

This study compared a specialized *DBT-AI chatbot* with a *general-purpose LLM chatbot* in a month-long deployment of a mobile mental health intervention. By analyzing longitudinal conversations using a six-dimensional behavioral framework, we found that the *DBT-AI chatbot* consistently provided clearer guidance, more actionable support, and stronger facilitation of reflection, leading to greater emotional improvement than the general-purpose chatbot. Among the behavioral dimensions, Guidance Clarity showed the strongest association with positive emotional outcomes. Our findings suggest that effective AI-supported mental health conversations rely not only on emotional validation but also on helping users engage in structured reflection and actionable coping strategies. These results highlight the value of grounding conversational AI in evidence-based therapeutic knowledge to deliver more consistent and skills-focused support. We hope this work provides practical insights for developing conversational mental health systems that are more effective, therapeutically grounded, and suitable for real-world deployment.

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